

WASP-10b

R-2410

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_100913 · created 2026-08-21T01:36:26+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2455452.68378 +/- 0.00078 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.211 +/- 0.062
Transit depth (Rp/Rs)^2	4.46 +/- 2.63 [%]
Orbital Inclination (inc)	89.93 +/- 1.87
Airmass coefficient 1 (a1)	0.4 +/- 0.22
Airmass coefficient 2 (a2)	0.58 +/- 0.39
Scatter in the residuals of the lightcurve fit is	62.33 %
Best Comparison Star	#1 - [286, 411]
Optimal Aperture	1.8
Optimal Annulus	14.48
Transit Duration (day)	0.0942 +/- 0.0094

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.211 ± 0.062 vs 0.1592 (deviation 0.84σ)
Duration fit vs published	0.0942 d vs 0.0946 d (deviation 0.04σ)
Mid-time O–C	-0.2 min
Depth SNR	3.4
Residual scatter	62.33 %

Light curve

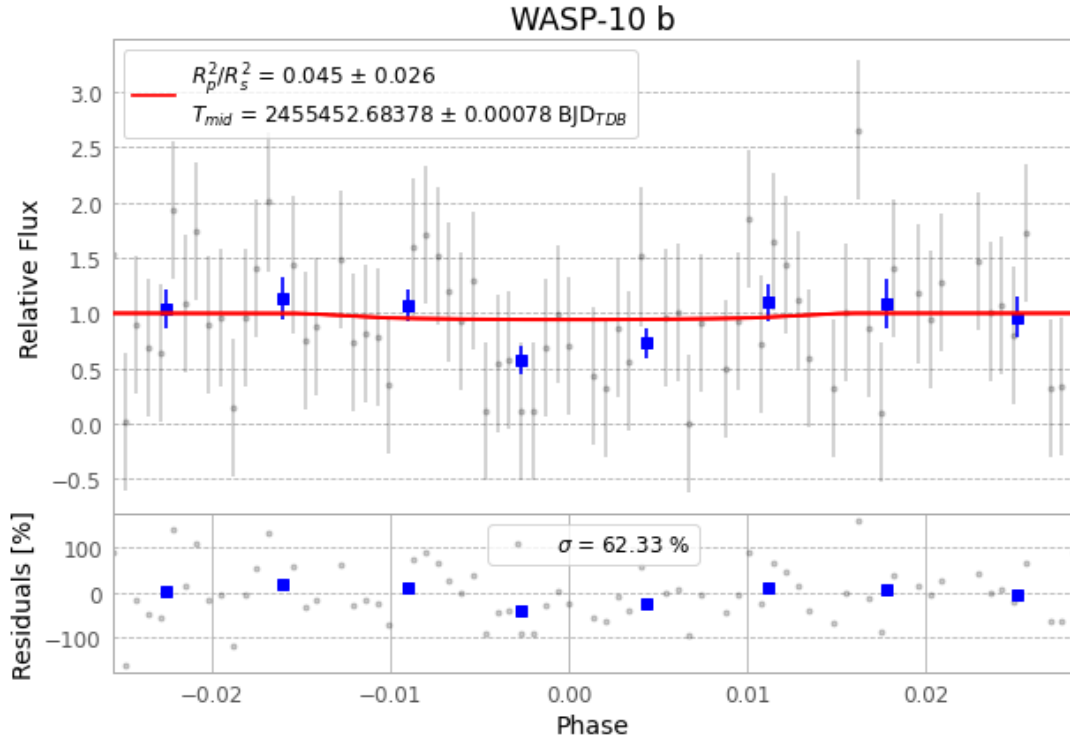


Figure 1 — FinalLightCurve_WASP-10 b_2010-09-12.png (SHA-256 e6d3f12c608ca7e4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [139, 402], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f9ddb308b60ddb32...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

82 FITS frames (54 MB), WASP-10100913021913.FITS → WASP-10100913062214.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-100913.log (a65e3dddf2ab...), FinalLightCurve_WASP-10 b_2010-09-12.png (e6d3f12c608c...), FinalLightCurve_WASP-10 b_2010-09-12.csv (3160df02cb51...)

Compute resources

Wall time 174.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 01:36Z.